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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IACUC: Ethics statement All experiments were approved by University of Illinois at Chicago (UIC) Institutional Animal Care and Use Committee (IACUC) and conformed to the guidelines set by United States Animal Welfare Act and the National Institutes of Health.
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
In vivo kinetic analysis of ICG-p28 MDA-MB-231 cells were injected s.c. into 4- to 5-week-old female athymic mice.
Subject Demographics
Age: Animal models Tumour models were established by subcutaneously (s.c.) implanting 1.0 x 106 MDA-MB-231, IOWA-1T or MDA-MB-231-iRFP cells into the right back (kinetic analysis) and mammary fat pads (MFPs qualitative margin analysis) of 4- to 5-week-old female athymic mice (Jackson Laboratory).
Randomization
Quantitative assessment of ICG-p28 for tumour margin determination and tumour recurrence follow-up Once NSG mice developed PDX tumours of similar sizes (200-300 mm3), the mice were randomly divided into five groups: ICG-p28, ICG alone, ICG-AA3H peptide, ICG-RI p28, and PBS.
Blinding
The animal surgeon was blinded to the experimental conditions such as animal identification and treatment group.
Power Analysis
Power analysis was done to determine appropriate animal size for the studies.
Replication

not required.
Cell Line Authentication
not detected.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental Models: Cell Lines			
Cell cultures The human breast cancer cell lines MDA-MB-231 (ATCC Cat: HTB-26, RRID: CVCL_0062) and IOWA-1T (ATCC Cat CRL-3292, RRID:CVCL_6G16) were obtained from American Type Culture Collection (Manassas, VA).	breast		(RRID:CVCL_0062)(link)
	IOWA-1T	ATCC	(ATCC Cat# CRL-3292, RRID:CVCL_6G16)(link)
MDA-MB-231-iRFP cells were established and characterized in our laboratory.	MDA-MB-231-iRFP		
Detection of cancer cells in vitro by ICG-p28 MDA-MB-231 cells were treated with 0.5 mgkg (130 nM) ICG-p28 followed by 2 h incubation at 37 degC in a humidified incubator with 5 CO2.	MDA-MB-231		Suggestion: RRID:CVCL_0062(link)
Experimental Models: Organisms/Strains			
For additional quantitative margin analysis, tumour models were established by implanting triple-negative PDX cells (stage IV) in the MFPs of 4- to 5-week-old female NODScidIL2Rgnull (NSG) mice (Jackson Laboratory).	NODScidIL2Rgnull (NSG)		
Recombinant DNA			
MDA-MB-231 cells were transfected with the piRFP plasmid with Lipofectamine 2000 (Invitrogen).	piRFP		Suggestion: RRID:Addgene_31857(link)
Software and Algorithms			
The tumour background ratio (TBR) was determined using ImageJ software (National Institutes of Health), where a region of interest (ROI) was drawn over the whole tumour and a background ROI was drawn over adjacent tissue.	ImageJ		Suggestion: ImageJ, RRID:SCR_003070(link)
Data processing and statistical analysis were performed using GraphPad Prism ver. 8 (GraphPad Software) and R version 4.0.5 (The R Foundation for Statistical Computing).	GraphPad Prism		
	GraphPad		Suggestion: GraphPad Prism, RRID:SCR_002798(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Friedmans nonparametric rank test was used for recurrent tumour volume data.	Friedmans nonparametric rank test
F test was used for Alu sequences after validating the normality for its log transformation as shown in Supplementary Figure S5.	F test
Oligonucleotides	
To identify human cells in the mouse tissues, primers specific for huAlu (sense: 5- ACG CCT GTA ATC CCA GCA CTT -3 and antisense: 5- TCG CCC AGG CTG GAG TGC A -3) were used to amplify the huAlu repeats present in the extracted genomic DNA by real-time PCR (5 ng genomic DNA, 0.5 uM each primer, and PowerUp SYBR Green Master Mix Life Technologies, USA).	5- ACG CCT GTA ATC CCA GCA CTT -3
	5- TCG CCC AGG CTG GAG TGC A -3
A quantitative measure of amplifiable mouse DNA was obtained through amplification of the mouse GAPDH genomic DNA sequence (mGAPDH) with mGAPDH-specific primers (sense: 5- AGG TCG GTG TGA ACG GAT TTG -3 antisense: 5- GGG GTC GTT GAT GGC AAC A -3) using the same PCR conditions described for huAlu.	5- AGG TCG GTG TGA ACG GAT TTG -3
	5- GGG GTC GTT GAT GGC AAC A -3

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq